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The program I created for HW-1 was a Java implementation of the board game Farkle. It was implemented with the intent of it being a singleplayer game, and the player only playing one round of Farkle at a time. The program starts in Farkle.java, where an instance of the StartGame class is called, which starts the game. Everything else happens in the StartGame class file, where it uses another file, DiceWork, in order to manipulate, mix, and move the dice throughout the game. It is also responsible for printing out the final table. DiceWork is responsible for randomly shuffling the dice, changing the meld throughout the game, printing the table, and computing the players score using the meld. The design also consists of varying private and public class variables and functions, which is intended in a way to better organize the class files and its purposes. Public variables/functions are used outside the class, and private variables/functions are used only in the specific class. I also implemented a lot of helper functions throughout DiceWork, which aid other functions in fulfilling their purposes (such as computeScore). Design issues I came across were implementing the computeScore function in DiceWork. This was very complicated because I wanted to keep the function clean and not just a bunch of if/else statements, as well as actually figuring out how to implement the scoring, which had me stumped for a while. I ended up deciding to compact the function a bit by implementing a helper function (checkForDuplicates), which helped me with the triple numbers and their scoring features, since I thought this was a better way to make my code easier to read, and more compact. I also took

advantage of 2D arrays, since they made it so I didn't have to do a lot of unnecessary assigning, and less if statements/for loops. However in the end, I realized that things I could've done differently if I had more time were not using as much for loops, and using less 2D arrays. The amount of for loops I used were a lot, and I realized that I could've made the program less redundant by cutting the amount of for loops I used. I also realized that the use of 2D arrays made sense in my head at the time, but later made my program more complicated than it should have been, which was also a pain to comment and explain later on.